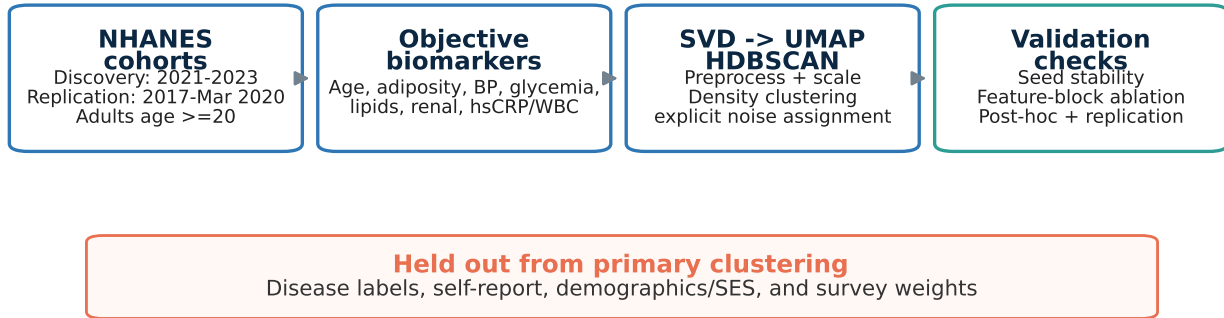


Leakage-controlled label-free phenotyping design

Biomarker-derived geometry first; clinical labels and survey context reserved for interpretation.



Interpretation boundary: descriptive biomarker-density tendencies only, not diagnostic classes or treatment rules.